Intro to Machine Learning
Quiz 3 Practice Problems
Practice problems, not turned in — Solutions

Directions: These practice problems cover the material for Quiz 3: linear discriminant analysis in Part 1, quadratic discriminant analysis in Part 2, and basis functions and splines in Part 3. They are not turned in, but the quiz draws on them, so work through each one before looking at the solutions. Problems adapted from ISL cite the exercise; see the book for the original wording.

Part 1: Linear Discriminant Analysis

Notation follows ISL 4.4. Classes are $k = 1, \dots, K$, $\pi_k = \Pr(Y = k)$ is the prior, $f_k(x)$ is the density of X in class k, and $p_k(x) = \Pr(Y = k \mid X = x)$ is the posterior. When fitting LDA by hand, use the pooled variance with divisor $n - K$.

1. **The LDA discriminant.** (ISL 4.8, Exercise 2; ISL 4.4.1–4.4.2, ESL 4.3.)

   (a) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma^2)$, with the same σ^2 in every class. Show that the Bayes classifier assigns x to the class with the largest

   $$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k.$$

   (b) With $K = 3$ and $p = 4$, how many parameters does LDA estimate (priors, means and the shared covariance)? How many coefficients does multinomial logistic regression estimate? The LDA posterior has the same form as a multinomial logistic regression, so what is LDA doing with the extra parameters?

   Solution.

   (a) The Bayes classifier picks the k with the largest posterior, $p_k(x) = \pi_k f_k(x) / \sum_l \pi_l f_l(x)$. The denominator doesn't depend on k, so it picks the largest $\pi_k f_k(x)$, or equivalently the largest log:

   $$\log \pi_k + \log f_k(x) = \log \pi_k - \log\left(\sqrt{2\pi}\,\sigma\right) - \frac{(x - \mu_k)^2}{2\sigma^2}$$

   $$= \log \pi_k - \log\left(\sqrt{2\pi}\,\sigma\right) - \frac{x^2}{2\sigma^2} + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2}.$$

   The terms $-\log(\sqrt{2\pi}\,\sigma)$ and $-x^2/(2\sigma^2)$ are the same for every class (because σ is shared), so they don't change which k is largest. Dropping them leaves $\delta_k(x)$.

(b) LDA: $K-1=2$ priors, $Kp=12$ means and $p(p+1)/2=10$ covariance entries, so 24 parameters. Logistic regression: $(K-1)(p+1)=10$. Both end with a posterior that has 10 free coefficients (the same form), but LDA also models how X is distributed, which logistic regression leaves alone. If that model is right, the extra structure lets LDA use every observation to estimate the coefficients, so it has lower variance. If it is wrong, the coefficients are biased (the LDA vs. logistic lab, Sections 2 and 3).

2. **LDA with one predictor.** (ISL 4.8, Exercise 7; ISL 4.4.2.) We want to predict whether a stock will issue a dividend this year (yes or no) from X, last year's percentage profit. Among companies that issued a dividend, X averaged $\bar{x}=10$; among those that didn't, $\bar{x}=0$. The variance of X in both groups is $\hat{\sigma}^2=36$, and 80% of companies issued dividends. Assume X is normal in each group. Find the probability that a company with $X=4$ issues a dividend this year.

   Solution. By Bayes' theorem, with the normal density's constant $1/(\sqrt{2\pi}\,\sigma)$ cancelling,

$$\hat{\Pr}(\text{dividend} \mid X=4)=\frac{0.8\,e^{-(4-10)^2/72}}{0.8\,e^{-(4-10)^2/72}+0.2\,e^{-(4-0)^2/72}}$$

$$=\frac{0.8\,e^{-0.5}}{0.8\,e^{-0.5}+0.2\,e^{-0.222}}=\frac{0.4852}{0.4852+0.1601}=0.752.$$

   Even though 4 is closer to the "no" mean of 0 than to 10, the 4-to-1 prior for "yes" wins.

3. **Fitting LDA by hand.** (ISL 4.4.2.) A training set has one predictor:

$$\text{class 1: } x=1,2,3; \qquad \text{class 2: } x=5,6,6,7.$$

   (a) Compute $\hat{\pi}_k$, $\hat{\mu}_k$ and the pooled variance $\hat{\sigma}^2$.

   (b) Find the LDA decision boundary. Which way did the unequal priors move it from the midpoint of the means?

   Solution.

   (a) $n=7$, $K=2$. $\hat{\pi}_1=3/7$, $\hat{\pi}_2=4/7$; $\hat{\mu}_1=2$, $\hat{\mu}_2=6$. The within-class sums of squares are $1+0+1=2$ and $1+0+0+1=2$, so $\hat{\sigma}^2=(2+2)/(7-2)=0.8$.

   (b) From Problem 1(a), $\delta_1(x)=2x/0.8-4/1.6+\log(3/7)$ and $\delta_2(x)=6x/0.8-36/1.6+\log(4/7)$. Setting them equal, $5x=20+\log(3/4)$, so $x^*=4-0.058=3.942$. Class 2 has the larger prior, so the boundary moves toward class 1, the rarer class, and class 2 claims a little more of the line.

Part 2: Quadratic Discriminant Analysis

QDA assumes $X \mid Y=k \sim N(\mu_k, \Sigma_k)$; naive Bayes assumes $f_k(x)=\prod_{j=1}^{p} f_{kj}(x_j)$.

4. **QDA is quadratic.** (ISL 4.8, Exercise 3; ISL 4.4.3, ESL 4.3.) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma_k^2)$, with a different variance in each class. Find the discriminant $\delta_k(x)$ the Bayes classifier maximizes, and show that the decision boundary between two classes is not linear in x unless $\sigma_1 = \sigma_2$.

   Solution. As for LDA, the Bayes classifier maximizes $\log \pi_k + \log f_k(x)$. Now

$$\log \pi_k + \log f_k(x) = \log \pi_k - \tfrac{1}{2}\log(2\pi) - \log \sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2}.$$

   Only $-\frac{1}{2}\log(2\pi)$ is the same for every class, so

$$\delta_k(x) = -\frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2} - \log \sigma_k + \log \pi_k.$$

   The boundary between classes 1 and 2 is $\delta_1(x) = \delta_2(x)$, and the x^2 coefficient of $\delta_1(x) - \delta_2(x)$ is $\frac{1}{2}(1/\sigma_2^2 - 1/\sigma_1^2)$. It is zero only if $\sigma_1 = \sigma_2$. Otherwise the boundary is the solution of a quadratic, so it can be two points, and the classification regions need not be intervals.

5. **QDA by hand: same means, different spreads.** (ISL 4.4.3.) Two equally likely classes have $X \mid Y = 1 \sim N(0, 1)$ and $X \mid Y = 2 \sim N(0, 4)$.

   (a) Find the Bayes decision rule. Where is each class predicted?

   (b) Now let $\pi_1 = 0.8$. How does the rule change?

   Solution.

   (a) From Problem 4 with $\mu_1 = \mu_2 = 0$, $\sigma_1 = 1$, $\sigma_2 = 2$ and equal priors (which cancel), $\delta_1(x) = -x^2/2$ and $\delta_2(x) = -x^2/8 - \log 2$. Predict class 1 when

$$-\frac{x^2}{2} > -\frac{x^2}{8} - \log 2 \iff \frac{3x^2}{8} < \log 2 \iff |x| < \sqrt{\tfrac{8\log 2}{3}} = 1.36.$$

   Class 1 is predicted on $(-1.36, 1.36)$ and class 2 on both sides of it. The boundary is two points, and class 2's region is two separate pieces.

   (b) The prior term $\log(\pi_1/\pi_2) = \log 4$ now favours class 1: predict class 1 when $3x^2/8 < \log 2 + \log 4 = \log 8$, that is $|x| < \sqrt{8\log 8/3} = 2.35$. Class 1's interval widens.

6. **LDA or QDA?** (ISL 4.8, Exercise 5; ISL 4.4.3, 4.5.) Answer each and explain.

   (a) If the Bayes decision boundary is linear, do we expect LDA or QDA to do better on the training set? On the test set?

   (b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to do better on the training set? On the test set?

   (c) In general, as n increases, do we expect the test accuracy of QDA relative to LDA to improve, decline or stay the same?

   (d) True or false: even if the Bayes boundary is linear, QDA will probably achieve a lower test error rate than LDA, because QDA is flexible enough to model a linear boundary.

(e) With $p = 10$ predictors and $K = 3$ classes, how many covariance parameters do LDA, QDA and Gaussian naive Bayes each estimate? How many observations does each class need for QDA's $\hat{\Sigma}_k$ to be invertible?

Solution.

(a) QDA on the training set: it is more flexible and can always match LDA's boundary, so it fits the training data at least as well. LDA on the test set: a linear boundary is right, so QDA's extra flexibility buys no reduction in bias, only extra variance.

(b) QDA on the training set (for the same reason as in (a)). On the test set, usually QDA, if n is large enough to estimate the Σ_k well: LDA's linear boundary is biased, and that bias doesn't shrink with n. With small n and large p, LDA can still win.

(c) Improve. QDA's disadvantage is variance, which shrinks as n grows; any advantage it has is lower bias, which doesn't.

(d) False. If the Bayes boundary is linear, LDA's model is correct, so QDA's extra parameters add variance and no reduction in bias. QDA *can* represent the linear boundary, but its estimate will not be exactly linear, and it pays for the extra parameters in test error.

(e) Each Σ has $p(p+1)/2 = 55$ distinct entries. LDA: 55 (one shared matrix). QDA: $3 \times 55 = 165$. Gaussian naive Bayes: $Kp = 30$ variances (only the diagonal of each Σ_k). All three also estimate $Kp = 30$ means. A sample covariance computed from n_k points has rank at most $n_k - 1$, so QDA needs $n_k \geq p + 1 = 11$ in *every* class.

Part 3: Basis Functions and Splines

Here $I(\cdot)$ is the indicator, equal to 1 when its condition holds and 0 otherwise, and $g^{(m)}$ is the mth derivative of g, with $g^{(0)} = g$.

7. **Penalizing different derivatives.** (ISL 7.9, Exercise 2; ISL 7.5.) A curve $\hat{g}$ is fitted to n points by

$$\hat{g} = \arg\min_g \Big(\sum_{i=1}^{n} \big(y_i - g(x_i)\big)^2 + \lambda \int \big[g^{(m)}(x)\big]^2 \, dx \Big).$$

Sketch an example of $\hat{g}$ for each of the following.

(a) $\lambda = \infty$, $m = 0$.

(b) $\lambda = \infty$, $m = 1$.

(c) $\lambda = \infty$, $m = 2$.

(d) $\lambda = \infty$, $m = 3$.

(e) $\lambda = 0$, $m = 3$.

Solution. With $\lambda = \infty$ any curve with a nonzero penalty has an infinite criterion, so $\hat{g}$ must have $g^{(m)} = 0$ everywhere. Among those curves, it is the one with the smallest RSS.

(a) $g^{(0)} = g = 0$: $\hat{g}(x) = 0$ for every x, whatever the data.

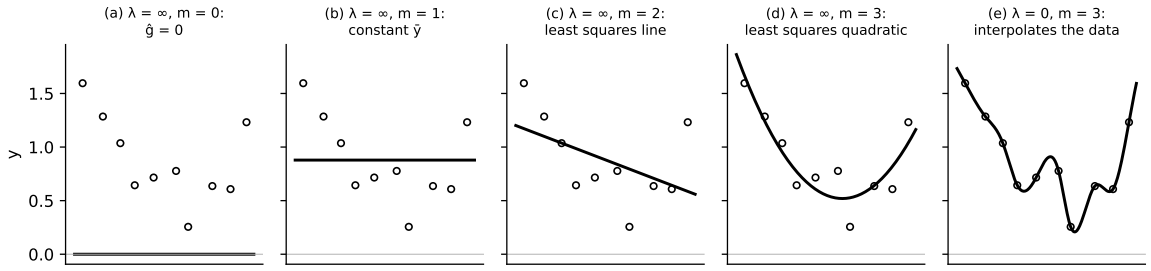
(b) $g' = 0$: g is a constant, and the constant that minimizes RSS is $\bar{y}$.

(c) $g'' = 0$: g is a line, so $\hat{g}$ is the least squares line.

(d) $g''' = 0$: g is a quadratic, so $\hat{g}$ is the least squares quadratic.

(e) With $\lambda = 0$ there is no penalty, so $\hat{g}$ only minimizes RSS. It passes through every point (RSS $= 0$), and between the points it can be as wiggly as needed. This is the smoothing spline with no smoothing: it interpolates the data and overfits.

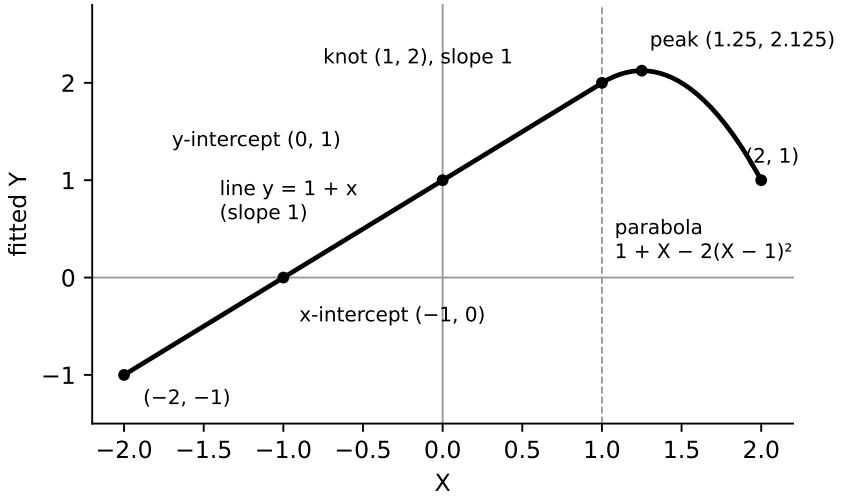
In general, $\lambda = \infty$ gives the least squares polynomial of degree $m - 1$.



8. **Sketching a fit from its basis functions, I.** (ISL 7.9, Exercise 3; ISL 7.3.) A curve is fitted with basis functions $b_1(X) = X$ and $b_2(X) = (X-1)^2 I(X \geq 1)$. The regression $Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + \varepsilon$ gives $\hat{\beta}_0 = 1$, $\hat{\beta}_1 = 1$ and $\hat{\beta}_2 = -2$. Sketch the fitted curve for $-2 \leq X \leq 2$, marking the intercepts, slopes and anything else of interest.

Solution. The fitted curve is $\hat{f}(X) = 1 + X - 2(X-1)^2 I(X \geq 1)$.

- For $X < 1$, $b_2(X) = 0$, so $\hat{f}(X) = 1 + X$: a line with slope 1, y-intercept $(0, 1)$ and x-intercept $(-1, 0)$. At $X = -2$ it is -1.
- For $X \geq 1$, $\hat{f}(X) = 1 + X - 2(X-1)^2$, a downward parabola. At $X = 1$ both pieces equal 2, and the parabola's slope $1 - 4(X-1)$ is 1, the same as the line's, so the curve has no jump and no kink at $X = 1$.
- The slope is 0 at $X = 1.25$, where the curve peaks at $\hat{f} = 2.125$. At $X = 2$, $\hat{f} = 1$.



9. **Sketching a fit from its basis functions, II.** (ISL 7.9, Exercise 4; ISL 7.2, 7.3.) A curve is fitted with basis functions

$$b_1(X) = I(0 \le X \le 2) - (X-1)I(1 \le X \le 2),$$

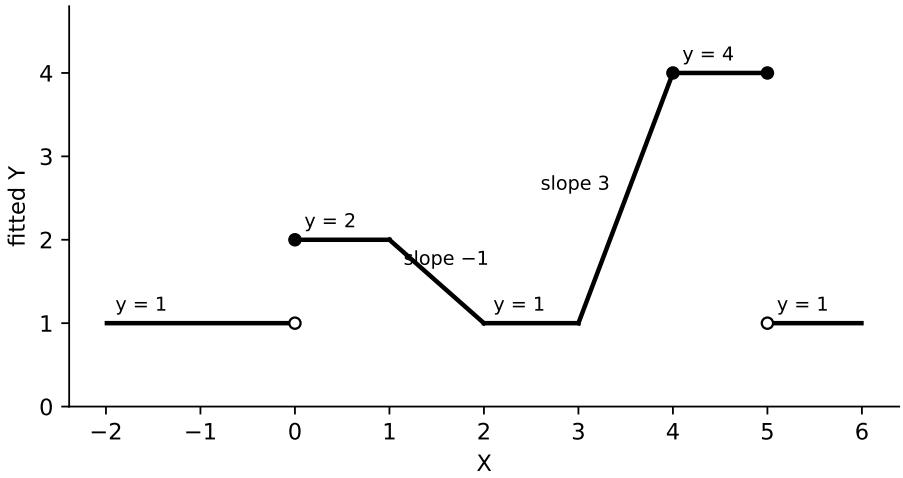
$$b_2(X) = (X-3)I(3 \le X \le 4) + I(4 < X \le 5).$$

The regression $Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + \varepsilon$ gives $\hat{\beta}_0 = 1$, $\hat{\beta}_1 = 1$ and $\hat{\beta}_2 = 3$. Sketch the fitted curve for $-2 \le X \le 6$, marking the intercepts, slopes and anything else of interest.

Solution. Work out each basis function piece by piece. $b_1(X) = 1$ on $[0, 1]$, $1-(X-1) = 2-X$ on $[1, 2]$, and 0 elsewhere. $b_2(X) = X - 3$ on $[3, 4]$, 1 on $(4, 5]$, and 0 elsewhere. Then $\hat{f}(X) = 1 + b_1(X) + 3b_2(X)$:

Interval	$\hat{f}(X)$	Shape
$-2 \le X < 0$	1	flat
$0 \le X \le 1$	2	flat
$1 \le X \le 2$	$3 - X$	slope -1, from 2 down to 1
$2 < X < 3$	1	flat
$3 \le X \le 4$	$1 + 3(X-3)$	slope 3, from 1 up to 4
$4 < X \le 5$	4	flat
$5 < X \le 6$	1	flat

The y-intercept is $\hat{f}(0) = 2$. The curve jumps up from 1 to 2 at $X = 0$ and down from 4 to 1 just after $X = 5$, and is continuous everywhere else (at $X = 1$, 2, 3 and 4 the pieces meet).



10. **Penalizing the third or the fourth derivative?** (ISL 7.9, Exercise 5; ISL 7.5.) Consider

$$\hat{g}_1 = \arg\min_g \Big(\sum_{i=1}^n \big(y_i - g(x_i)\big)^2 + \lambda \int \big[g^{(3)}(x)\big]^2 \, dx \Big),$$

$$\hat{g}_2 = \arg\min_g \Big(\sum_{i=1}^n \big(y_i - g(x_i)\big)^2 + \lambda \int \big[g^{(4)}(x)\big]^2 \, dx \Big).$$

(a) As $\lambda \to \infty$, which of $\hat{g}_1$ and $\hat{g}_2$ has the smaller training RSS?

(b) With $\lambda = 0$, which has the smaller training RSS and test RSS?

Solution.

(a) $\hat{g}_2$. As in Problem 7, $\lambda \to \infty$ forces $g^{(3)} = 0$ for $\hat{g}_1$, so it becomes the least squares quadratic, and $g^{(4)} = 0$ for $\hat{g}_2$, so it becomes the least squares cubic. Every quadratic is a cubic, so the best cubic fits the training data at least as well: $\hat{g}_2$'s training RSS is smaller (or equal).

(b) The same. With $\lambda = 0$ the penalty is gone, and both criteria are just the RSS. Both curves interpolate the training data, so both have training RSS 0, and both overfit, so neither has an advantage in test RSS. (Both test RSS will typically be large.)